

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/ CO)	Marks
Q. 1 Solve <u>any two</u> of the following.		12
A) Define the terms 'soil', 'soil mechanics' and soil engineering. What are limitations of soil engineering?	CO 1	06
B) A sample of dry soil weighs 68 gm. Find the volume of voids if the total volume of the sample is 40 ml and the specific gravity of solids is 2.65. Also determine the voids ratio.	CO 1	06
C) A soil has a porosity of 40 %, the specific gravity of solids of 2.65 and a water content of 12 %. Determine the mass of water required to be added to 100 m ³ of this soil for full saturation.	CO 1	06
Q.2 Solve <u>any two</u> of the following.		12
A) What is relative density? How it is determined? What is the importance for coarse-grained soil?	CO 1	06
B) A sample of sand has a volume of 1000 ml in its natural state. Its minimum volume when compacted is 840 ml. When gently poured in a measuring cylinder, its maximum volume is 1370 ml. Determine the relative density.	CO 1	06
C) The liquid limit and plastic limit of a clay are 100% and 25%, respectively. From a hydrometer analysis it has been found that the clay soil consists of 50% of particles smaller than 0.002 mm. Indicate the activity classification of this clay and the probable type of clay mineral.	CO1	06
Q. 3 Solve <u>any two</u> of the following.		12
A) Enlist and explain in detail the factors affecting permeability of soils.	CO 2	06
B) Determine the average coefficient of permeability in the horizontal and vertical directions for a deposit consisting of three layers of thickness 5 m, 1 m and 2.5 m and having the coefficients of permeability of 3×10^{-2} mm/sec, 3×10^{-5} mm/sec, and 4×10^{-2} mm/sec, respectively. Assume the layers are isotropic.	CO 2	6
C) A glass cylinder 5 cm internal diameter and with a screen at the bottom was used as a falling head permeameter. The thickness of the sample was 10 cm. With the water level in the tube at the start of the test as 50 cm above the tail water, it dropped by 10 cm in one minute, the tail water level remaining unchanged. Calculate the value of k for the sample of the soil. Comment on the nature of the soil.	CO2	06

Q.4 Solve **any two** of the following. 12

- A) Enlist the types of shear test based on drainage conditions & explain in detail. CO2 06
- B) A series of direct shear tests was conducted on a soil, each test was carried out till the sample failed. The following results were obtained. CO2 06

Sample No.	Normal stress (kN/m ²)	Shear stresses (kN/m ²)
1	15	18
2	30	25
3	45	32

Determine the cohesion intercept and angle of shearing resistance.

- C) The following results were obtained from a series of consolidated undrained tests on a soil, in which the pore water pressure was not determined. CO2 06

Sample No.	Confining pressure (kN/m ²)	Deviator stress at failure (kN/m ²)
1	100	600
2	200	750
3	300	850

Determine the cohesion intercept and angle of shearing resistance.

Q. 5 Solve **any two** of the following. 12

- A) Enlist and explain in detail the factors affecting the compaction. CO 2 06
- B) An earth embankment is compacted at a water content of 18% to a bulk density of 19.2 kN/m³. If the specific gravity of the sand is 2.7, find the void ratio and the degree of saturation of the compacted embankment. CO 2 06
- C) A clay stratum 5 m thick has the initial void ratio of 1.50 and the effective overburden pressure of 120 kN/m². When the sample is subjected to an increase of pressure of 120 kN/m², the void ratio reduces to 1.44. Determine the coefficient of the volume compressibility and the final settlement of the stratum. CO 2 06

***** End *****